

Supplementary Information

Document caption: Detailed methods, supplementary analyses, and results including p-values.

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1 Current climate conditions

Data is available 2007 – 2022. 2008 is the first whole year of temperature data. Data available from <https://data.g-e-m.dk> ([GEM 2026f](#), [2026d](#)).

1-1 Temperature

Monthly mean temperature

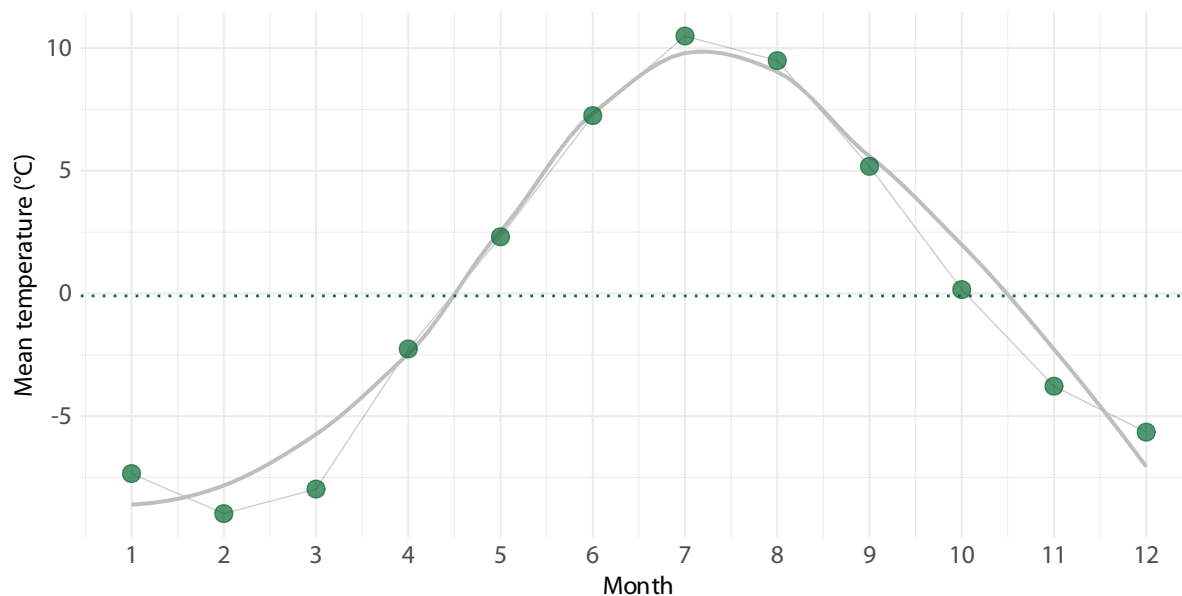


Figure 1: Mean temperature pr month. Mean air temperature (°C) pr month, based on data from 2007 to 2022. The dashed line represent yearly mean of -0.1°C.

The mean annual temperature is -0.1 °C. The warmest months are July (10.5 °C), August (9.5 °C), June (7.2 °C). The coldest months are January (-7.3 °C), March (-8 °C), February (-9 °C).

Yearly mean temperature

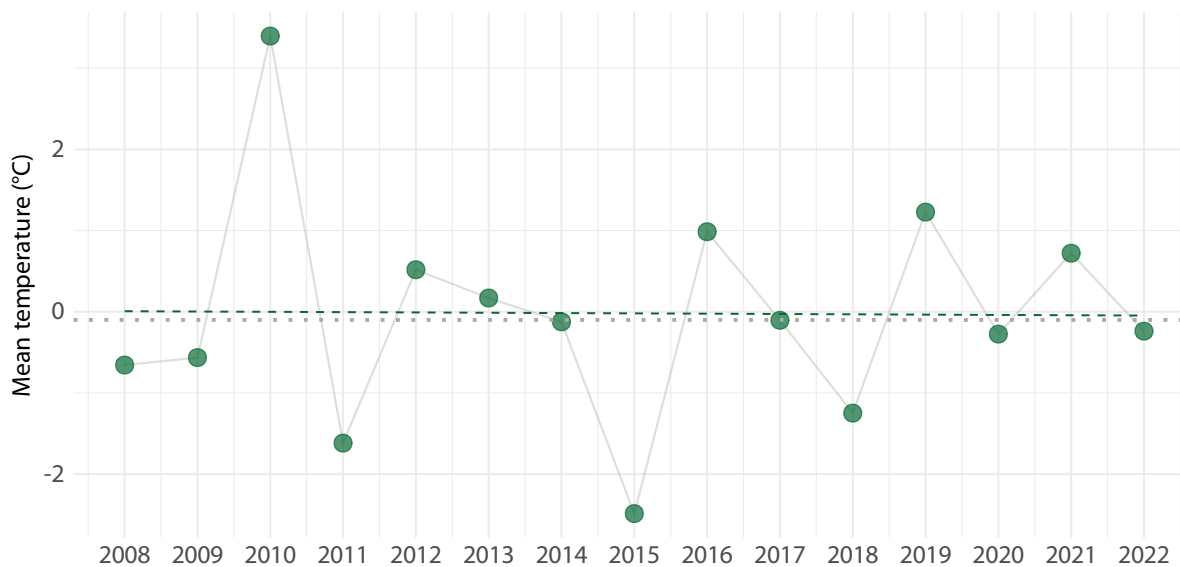


Figure 2: Mean annual air temperature pr year (°C) from 2008 to 2022 in Kangerluassunguaq. Data from 2007 is excluded because data is only from October, November and December. Gray dotted line indicate over all mean of -0.1003°C. Dashed line is model prediction.

Five year intervals of temperature

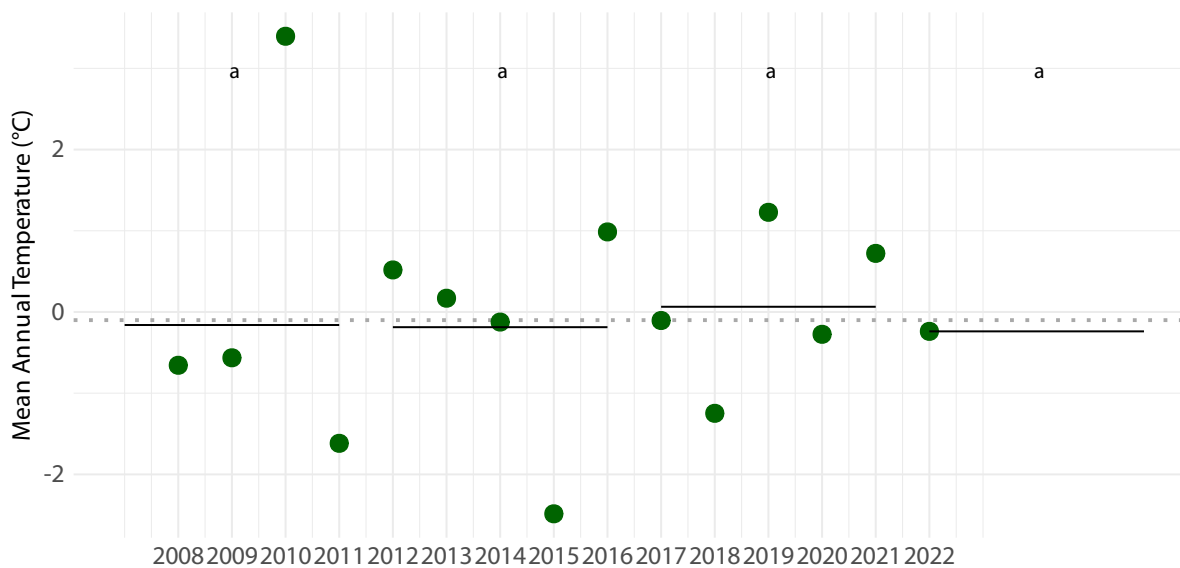


Figure 3: Five year intervals in temperature means. Shared letters indicate non-significant differences among estimated means according to compact letter display.

1-2 Precipitation

Data available from <https://data.g-e-m.dk> (GEM 2026d).

The mean annual precipitation is 886.15 mm.

The months with the most precipitation are September (138.64 mm), October (99.38 mm), August (98.28 mm), which account for 336.3 mm or 38 % of the yearly precipitation.

The months with the least precipitation are May (51.51 mm), February (37.62 mm), March (36.44 mm).

Monthly mean precipitation

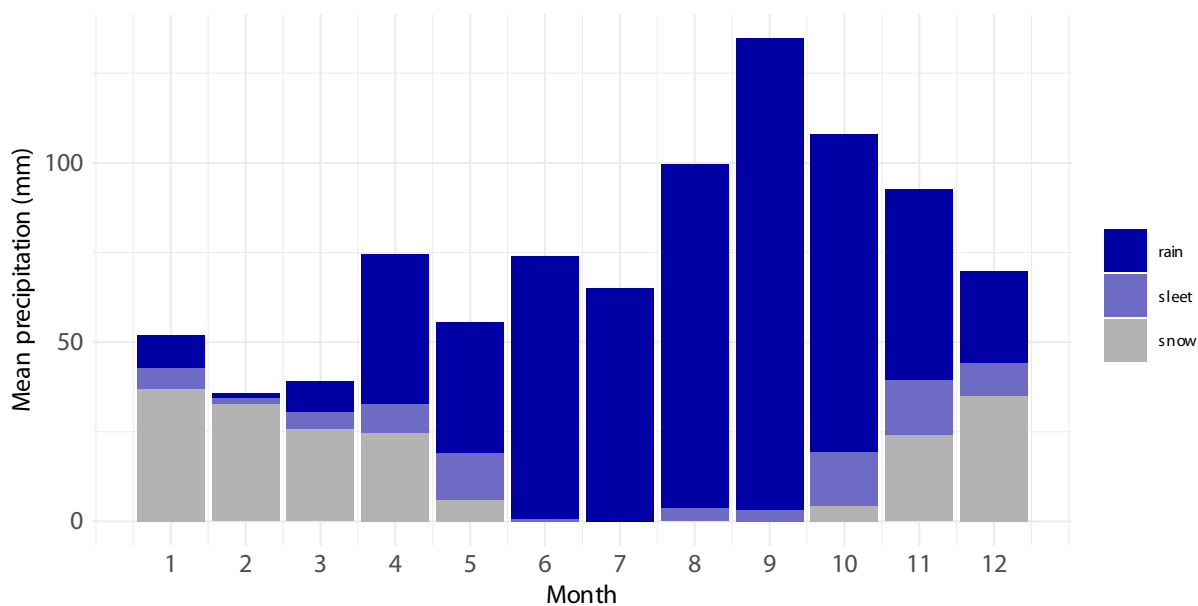


Figure 4: Mean precipitation type pr month. Mean monthly distribution of types of precipitation (mm). Based on data from 2008 to 2022 (requires join with temperature data which it only available till 2022). Sleet is defined as precipitation that fell when temperatures were between -1°C and 1°C .

Yearly mean precipitation

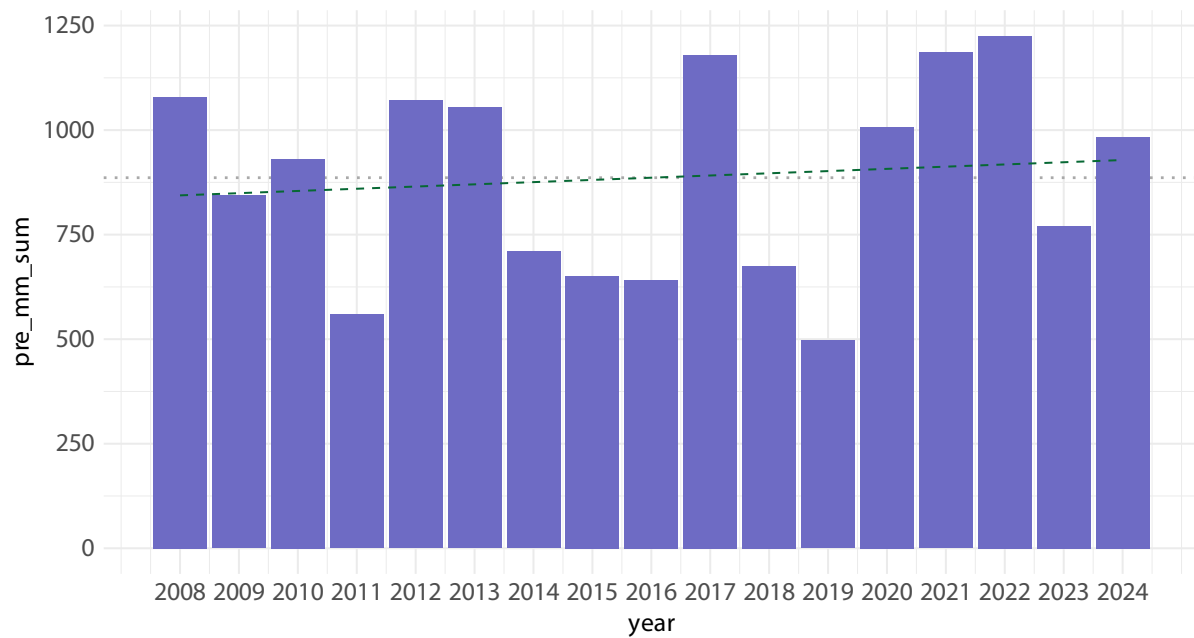


Figure 5: Mean yearly precipitation (mm) from 2008 to 2024. Data from 2007 is excluded because it was only from May - December. Blue dashed line indicate over all mean of 886.15 mm. Dashed line represent model prediction.

Five year interval precipitation

1-3 Radiation

Data available from <https://data.g-e-m.dk> (GEM 2026e)

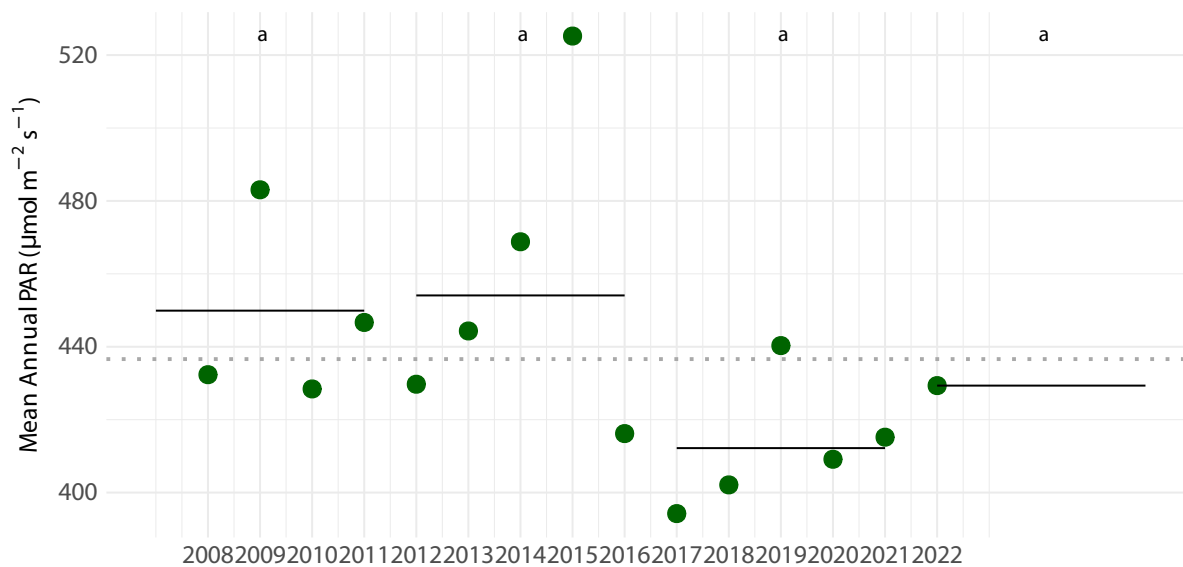


Figure 6: Five year intervals in PAR means. Shared letters indicate non-significant differences among estimated means according to compact letter display.

2 Historic climate data

2-1 Temperature

Data available from Cappelen ([2021](#)).

Station 4250 is Nuuk, and element 101 is precipitation.

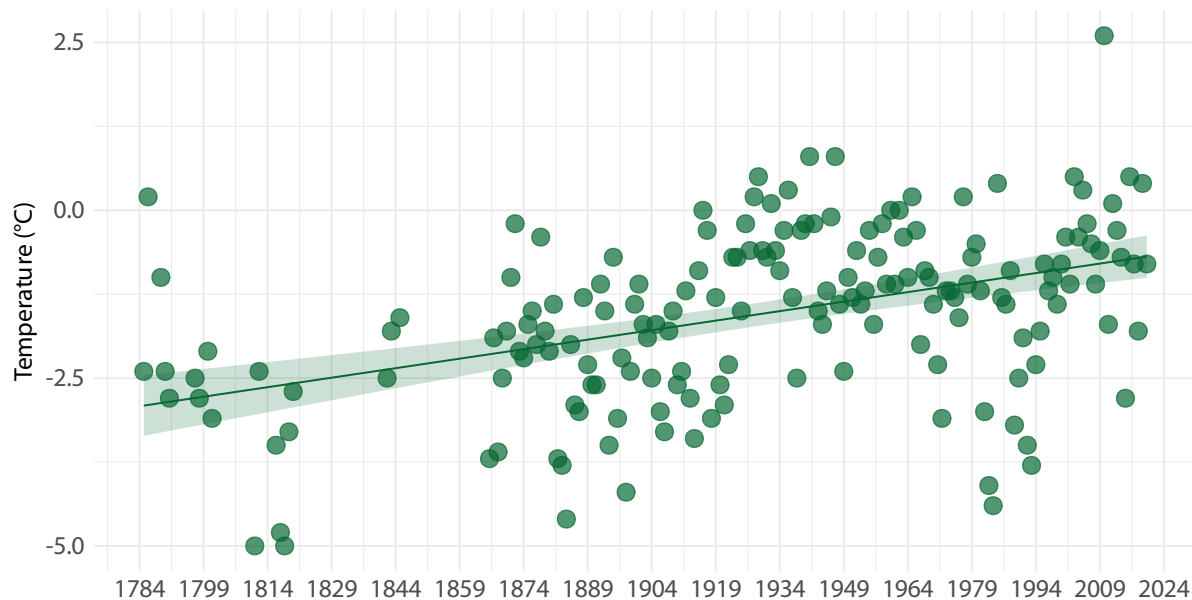


Figure 7: Historic yearly mean air temperature (°C) in Nuuk, from 1785 to 2020. Yearly mean of -1.6°C . Yearly mean ranges from -5°C to 2.6°C . Modelled mean temperature has increase from -2.92°C in 1784 to -0.65°C in 2024 (increase of 2.27°C or $0.095^{\circ}\text{C}/\text{decade}$.)

3 Statistical analysis

3-1 Diversity

Data available from <https://data.g-e-m.dk> (GEM 2026a).

Data has been filtered to remove sections with only one remaining plot and subsections in saltmarsh, due to too few replicates in this community type. Final calculates included 134 subsections (full transect consists of 162 subsection).

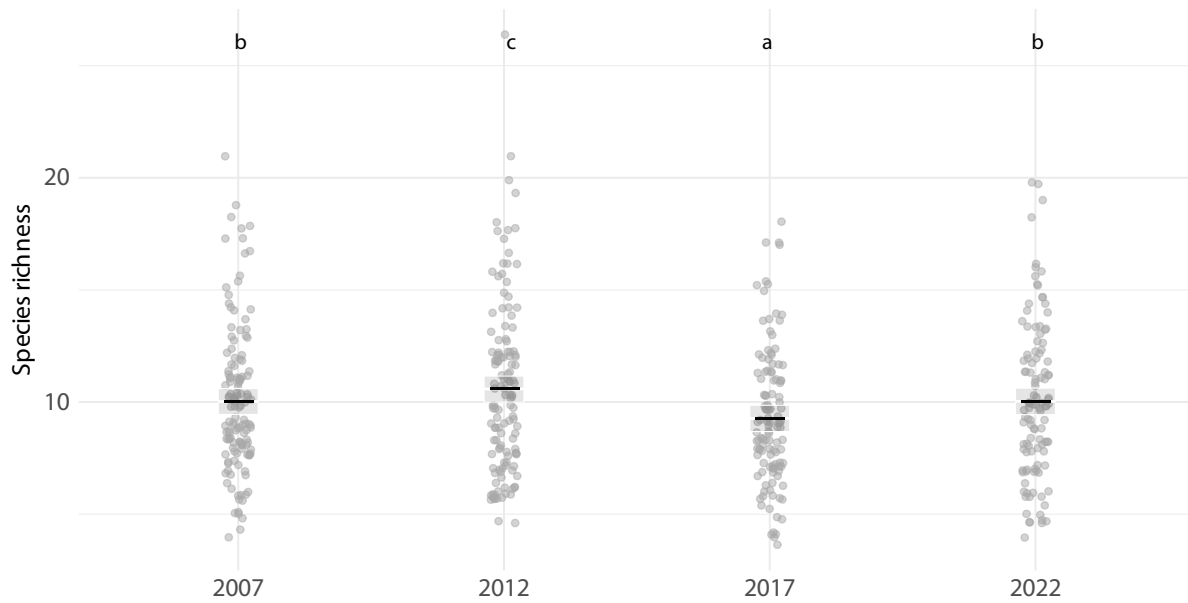
Richness

Figure 8: Species richness, number of species pr. subsection. Shared letters indicate non-significant differences among estimated means according to compact letter display.

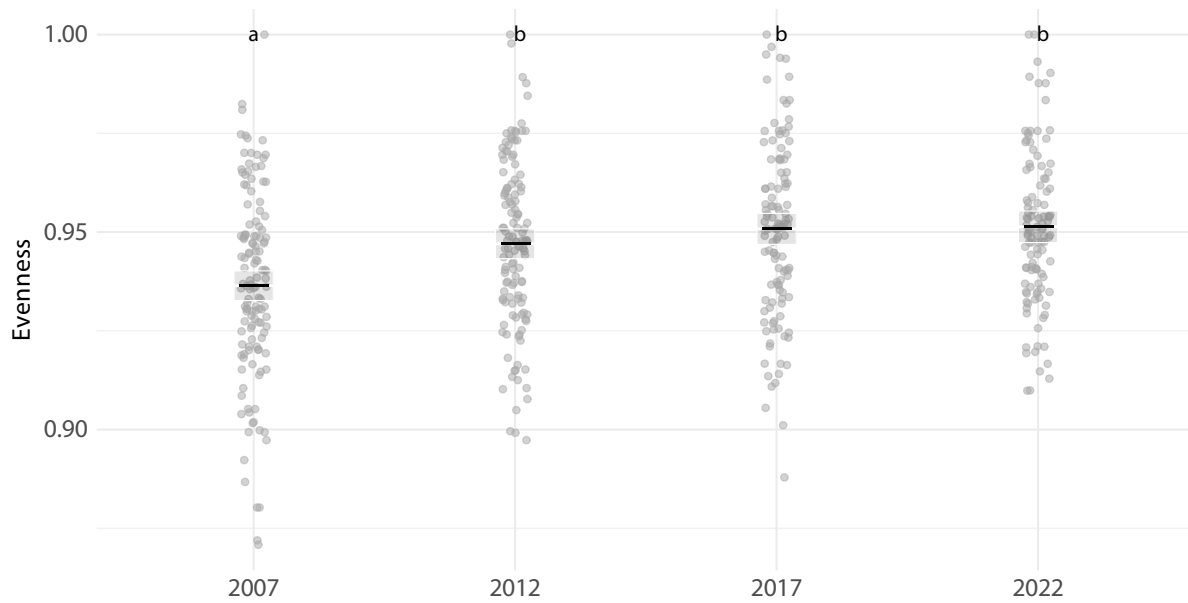
Evenness

Figure 9: Evenness pr. subsection. Shared letters indicate non-significant differences among estimated means according to compact letter display.

Shannon

Figure 10: Shannon diversity pr. year calculate pr subsection. Shared letters indicate non-significant differences among estimated means according to compact letter display.

Turnover

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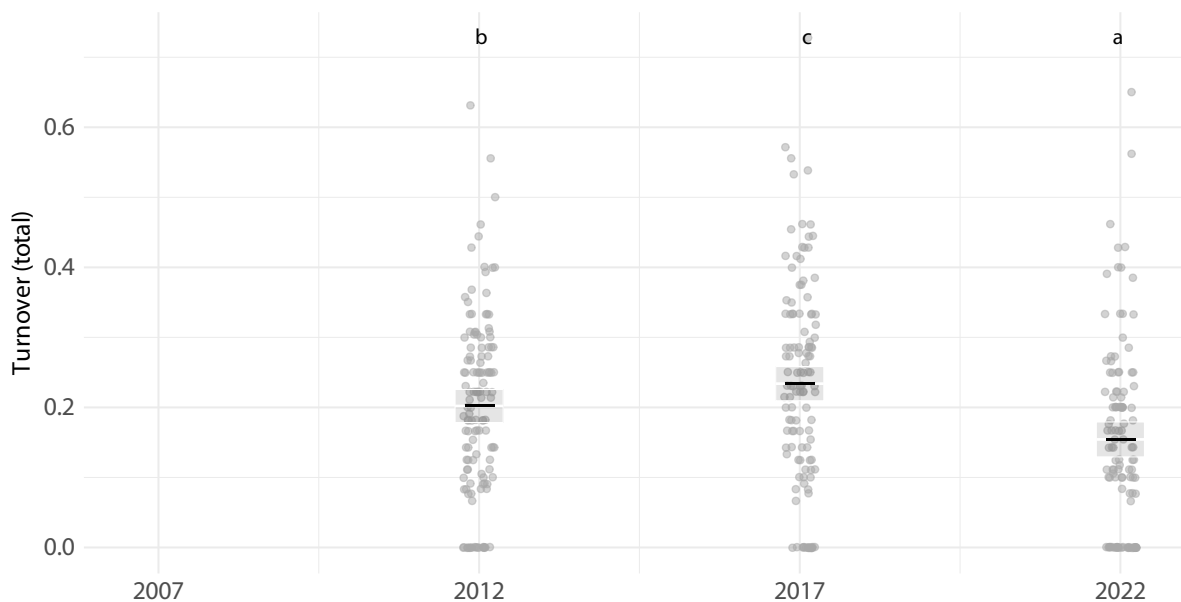


Figure 11: Turnover pr. year calculate pr subsection, but only pr present all survey years. A total of 131 subsections maintain more than 1 plots for all 4 surveys. Shared letters indicate non-significant differences among estimated means according to compact letter display.

Plots combined

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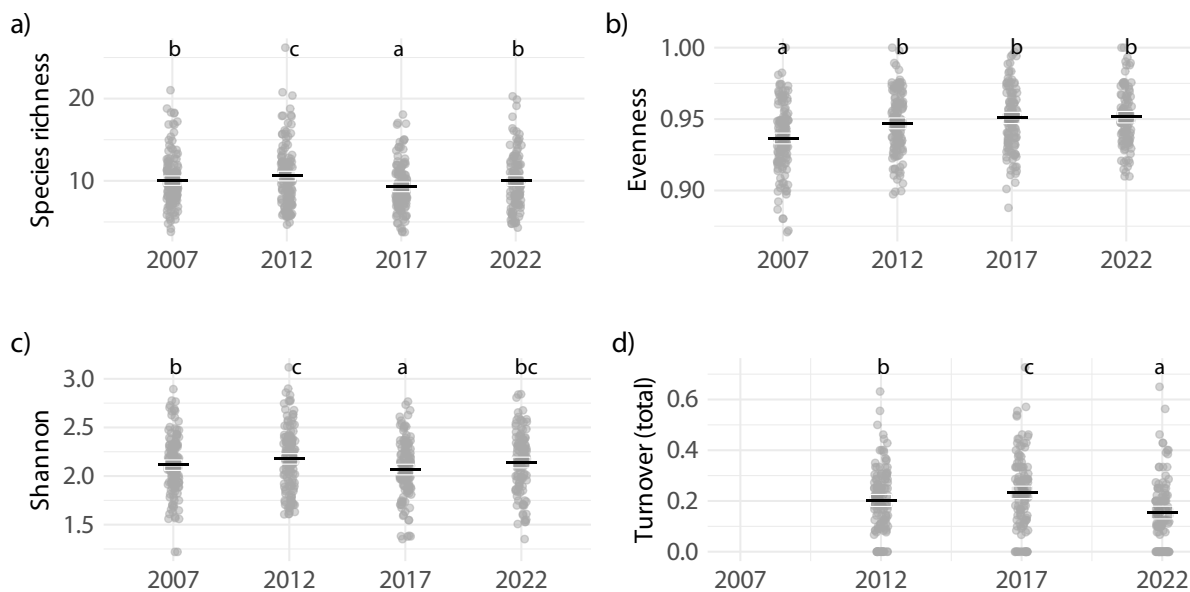


Figure 12: Diversity metrics, a) Shannon diversity, b) Species richness, c) Evenness, c) Species turnover. Shared letters indicate non-significant differences among estimated means according to compact letter display.

3-2 Community composition

NMDS

Community composition data ready for analysis consists of 134 subsections from 4 years of surveys. 99 species have been observed across all surveys.

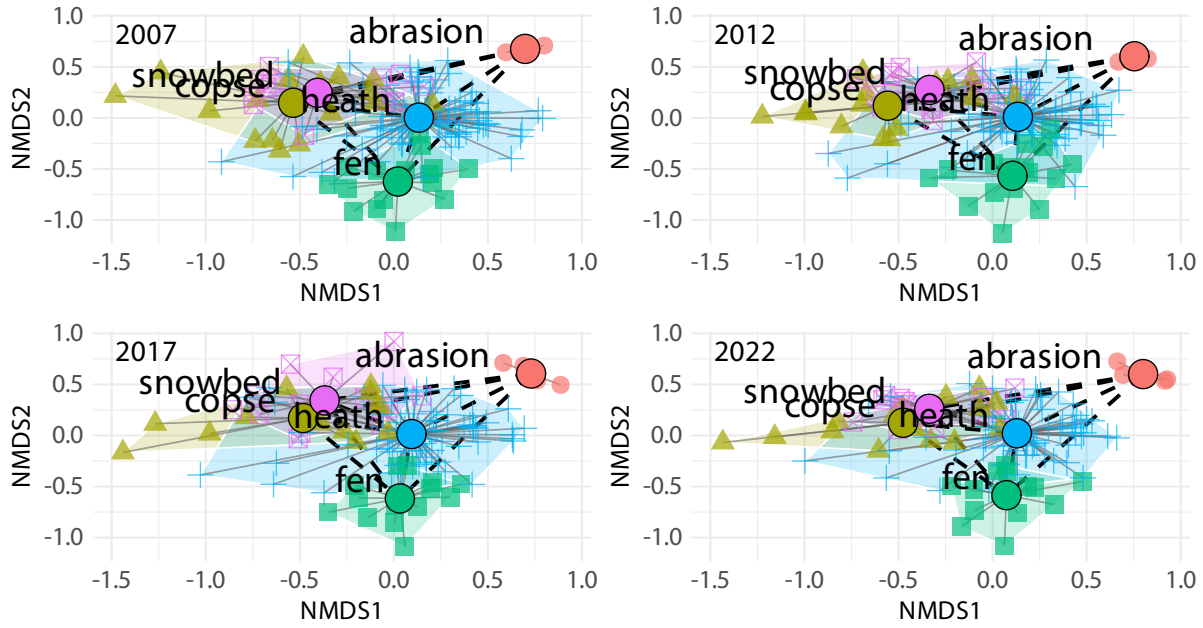


Figure 13: Plant community dynamics. Non-metric multidimensional scaling (NMDS) ordination of vegetation composition across years and vegetation types. Each panel represents subset of years of the full NMDS including all data to keep NMDS distances comparable. Grey lines connect plots within vegetation types, the within-type centroid distances. Black dashed lines indicate distances between vegetation-type centroids.

Between vegetation composition

Model: $\text{dist} \sim \text{factor}(\text{year}) + (1 \mid \text{veg_pair})$, fitted by REML

N observations: 40 | N groups (veg_pair): 10

Table 1: Fixed effects: centroid distance $\sim$ survey year (factor)

term	estimate	std.error	statistic
(Intercept)	0.886	0.119	7.462
Year 2012	-0.014	0.017	-0.780
Year 2017	-0.018	0.017	-1.024
Year 2022	-0.025	0.017	-1.448

Table 2: Estimated mean centroid distances per survey

year	emmean	SE	lower.CL	upper.CL
2007	0.886	0.119	0.618	1.154
2012	0.873	0.119	0.605	1.140
2017	0.868	0.119	0.600	1.136
2022	0.861	0.119	0.593	1.129

Table 3: Pairwise contrasts between surveys (Sidak-adjusted)

contrast	estimate	SE	t.ratio	p.value
year2007 - year2012	0.014	0.017	0.780	0.970
year2007 - year2017	0.018	0.017	1.024	0.897
year2007 - year2022	0.025	0.017	1.448	0.647
year2012 - year2017	0.004	0.017	0.244	1.000
year2012 - year2022	0.012	0.017	0.668	0.986
year2017 - year2022	0.007	0.017	0.424	0.999

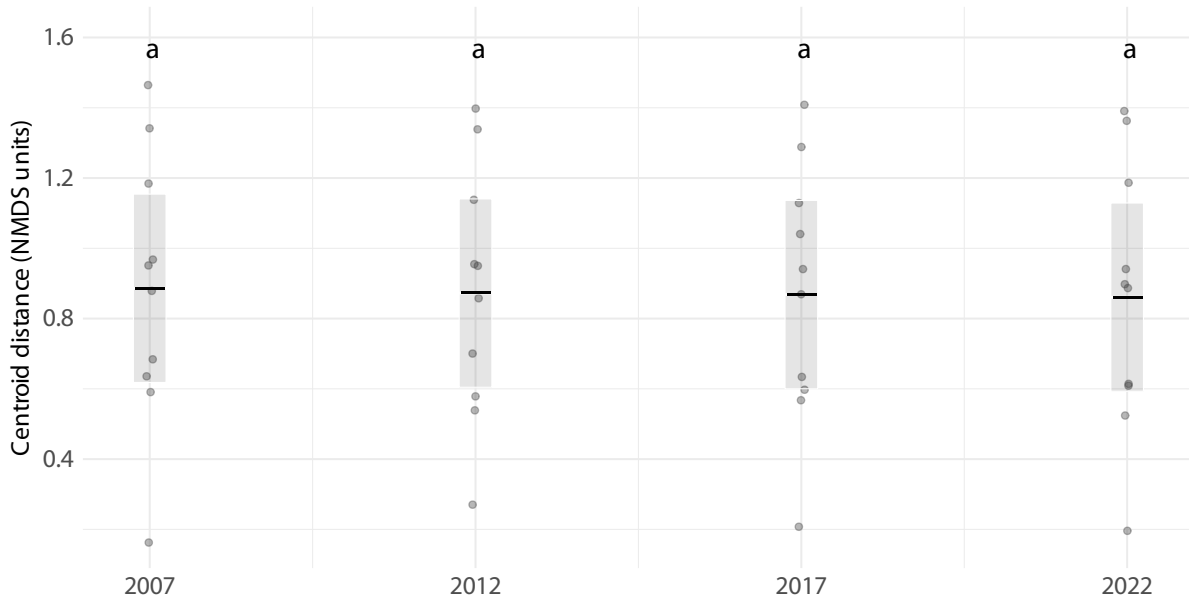


Figure 14: Between plant community dynamics. Estimated mean distance between vegetation-type centroids within each survey year based on linear mixed-effects models. Black lines are estimated means ($\pm$ SE), and grey dots represent raw subsection-level distances (NMDS units). Shared letters indicate non-significant differences among estimated means according to compact letter display. (Figure 5)

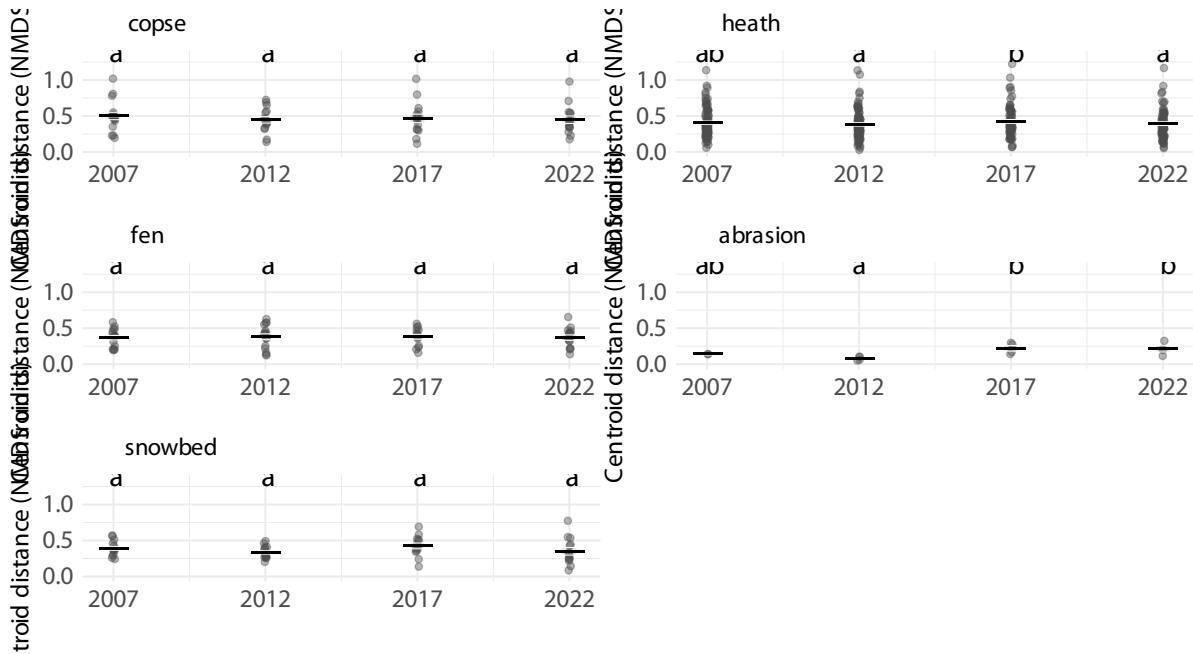
Within vegetation composition

Figure 15: Within community dynamics. Estimated mean centroids-subsection distances per community each survey year based on linear mixed-effects models. Black lines are estimated means ($\pm$ SE), and grey dots represent raw distances (NMDS units). Shared letters indicate non-significant differences among estimated means according to compact letter display. (Figure 6)

3-4 Plant functional types**Functional groups**

Table 4: Fixed effects per functional type

func_type	term	estimate	std.error	statistic
forb	(Intercept)	0.268	0.029	9.224
forb	Year 2012	0.082	0.033	2.502
forb	Year 2017	0.004	0.034	0.114
forb	Year 2022	0.038	0.034	1.098
graminoid	(Intercept)	0.469	0.028	16.860
graminoid	Year 2012	0.078	0.038	2.022
graminoid	Year 2017	0.054	0.040	1.367

func_type	term	estimate	std.error	statistic
graminoid	Year 2022	0.080	0.039	2.025
lichen	(Intercept)	0.491	0.035	14.063
lichen	Year 2012	0.141	0.021	6.754
lichen	Year 2017	-0.047	0.021	-2.216
lichen	Year 2022	0.050	0.022	2.317
shrub	(Intercept)	0.639	0.032	19.938
shrub	Year 2012	0.065	0.046	1.404
shrub	Year 2017	0.052	0.047	1.098
shrub	Year 2022	0.068	0.048	1.441
bryophyte	(Intercept)	0.886	0.016	55.412
bryophyte	Year 2012	0.062	0.016	3.940
bryophyte	Year 2017	-0.009	0.016	-0.597
bryophyte	Year 2022	0.062	0.016	3.870

Table 5: Estimated marginal means per functional type and survey year

func_type	year	emmean	SE	lower.CL	upper.CL	.group
forb	2007	0.268	0.029	0.211	0.326	a
forb	2012	0.351	0.028	0.295	0.407	a
forb	2017	0.272	0.030	0.213	0.331	a
forb	2022	0.306	0.030	0.247	0.365	a
graminoid	2007	0.469	0.028	0.415	0.524	a
graminoid	2012	0.547	0.028	0.491	0.603	a
graminoid	2017	0.523	0.030	0.465	0.582	a
graminoid	2022	0.549	0.030	0.491	0.608	a
lichen	2007	0.491	0.035	0.422	0.560	ab
lichen	2012	0.632	0.035	0.563	0.701	c
lichen	2017	0.444	0.035	0.375	0.513	a
lichen	2022	0.541	0.035	0.471	0.611	b
shrub	2007	0.639	0.032	0.576	0.702	a
shrub	2012	0.703	0.033	0.639	0.768	a
shrub	2017	0.691	0.035	0.622	0.760	a
shrub	2022	0.707	0.035	0.638	0.776	a
bryophyte	2007	0.886	0.016	0.854	0.917	a
bryophyte	2012	0.947	0.016	0.916	0.979	b
bryophyte	2017	0.876	0.016	0.844	0.908	a
bryophyte	2022	0.948	0.016	0.915	0.980	b

Table 6: Pairwise contrasts per functional type (Holm-adjusted)

func_type	contrast	estimate	SE	t.ratio	p.value
forb	year2007 - year2012	-0.082	0.033	-2.502	0.075
forb	year2007 - year2017	-0.004	0.034	-0.114	0.909
forb	year2007 - year2022	-0.038	0.034	-1.098	0.817
forb	year2012 - year2017	0.078	0.034	2.330	0.100
forb	year2012 - year2022	0.044	0.034	1.314	0.757
forb	year2017 - year2022	-0.034	0.035	-0.973	0.817
graminoid	year2007 - year2012	-0.078	0.038	-2.022	0.259
graminoid	year2007 - year2017	-0.054	0.040	-1.367	0.687
graminoid	year2007 - year2022	-0.080	0.039	-2.025	0.259
graminoid	year2012 - year2017	0.024	0.040	0.590	1.000
graminoid	year2012 - year2022	-0.002	0.040	-0.056	1.000
graminoid	year2017 - year2022	-0.026	0.041	-0.631	1.000
lichen	year2007 - year2012	-0.141	0.021	-6.754	0.000
lichen	year2007 - year2017	0.047	0.021	2.216	0.042
lichen	year2007 - year2022	-0.050	0.022	-2.317	0.042
lichen	year2012 - year2017	0.188	0.021	8.893	0.000
lichen	year2012 - year2022	0.091	0.021	4.256	0.000
lichen	year2017 - year2022	-0.097	0.022	-4.491	0.000
shrub	year2007 - year2012	-0.065	0.046	-1.404	0.899
shrub	year2007 - year2017	-0.052	0.047	-1.098	1.000
shrub	year2007 - year2022	-0.068	0.048	-1.441	0.899
shrub	year2012 - year2017	0.013	0.048	0.260	1.000
shrub	year2012 - year2022	-0.004	0.048	-0.079	1.000
shrub	year2017 - year2022	-0.016	0.050	-0.330	1.000
bryophyte	year2007 - year2012	-0.062	0.016	-3.940	0.000
bryophyte	year2007 - year2017	0.009	0.016	0.597	1.000
bryophyte	year2007 - year2022	-0.062	0.016	-3.870	0.000
bryophyte	year2012 - year2017	0.071	0.016	4.489	0.000
bryophyte	year2012 - year2022	-0.001	0.016	-0.038	1.000
bryophyte	year2017 - year2022	-0.072	0.016	-4.440	0.000

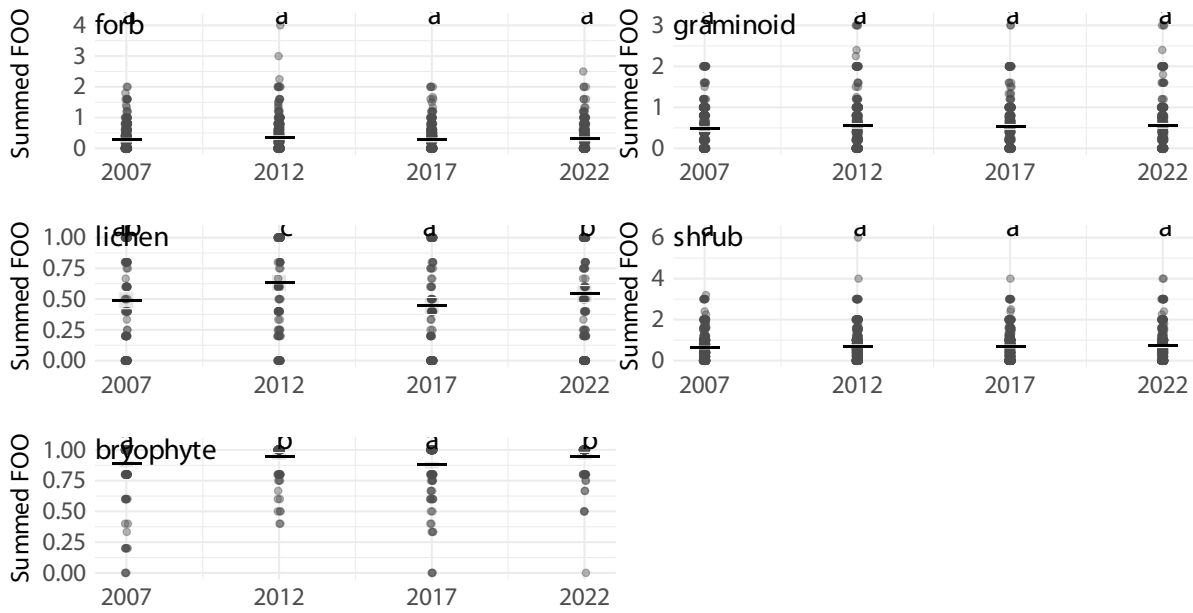


Figure 16: Plant functional group dynamics. Summed frequency of occurrence for plant functional groups. Shared letters indicate non-significant differences among estimated means according to compact letter display.

Individual species

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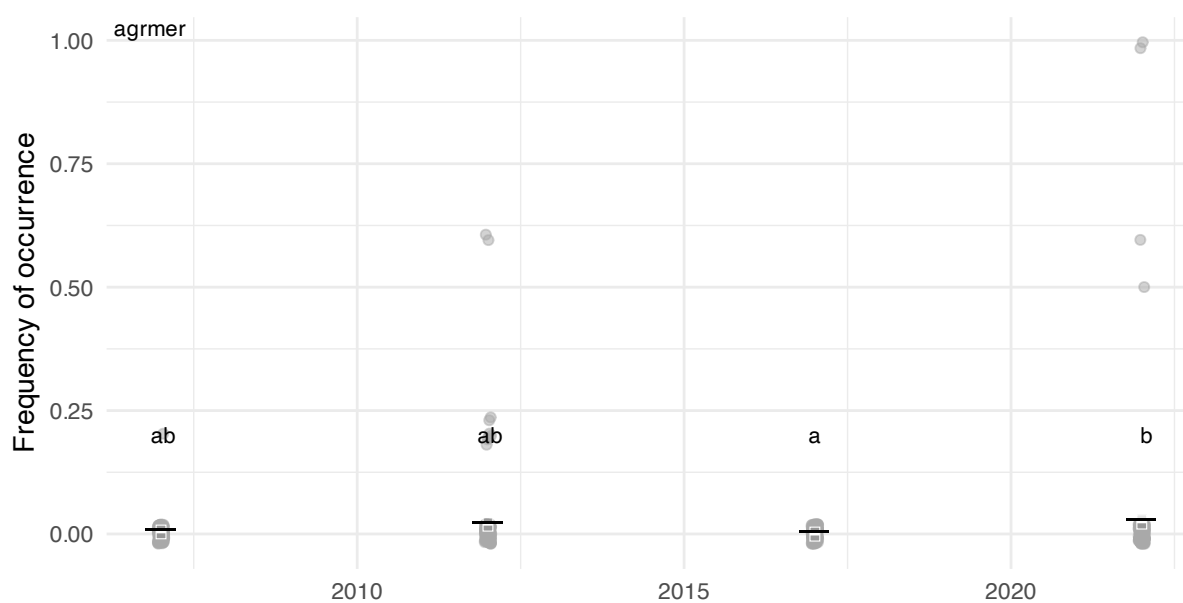
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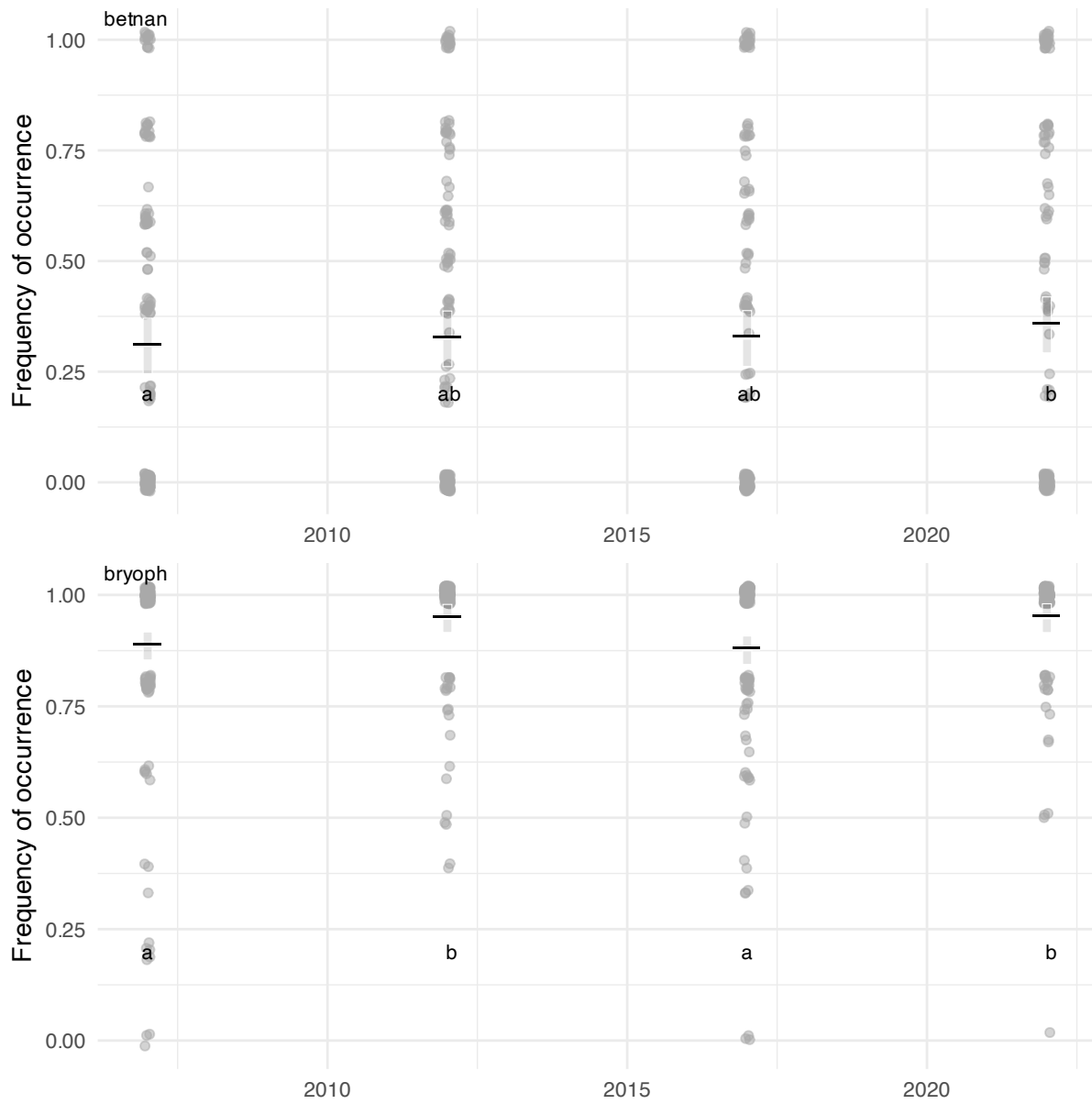
boundary (singular) fit: see help('isSingular')

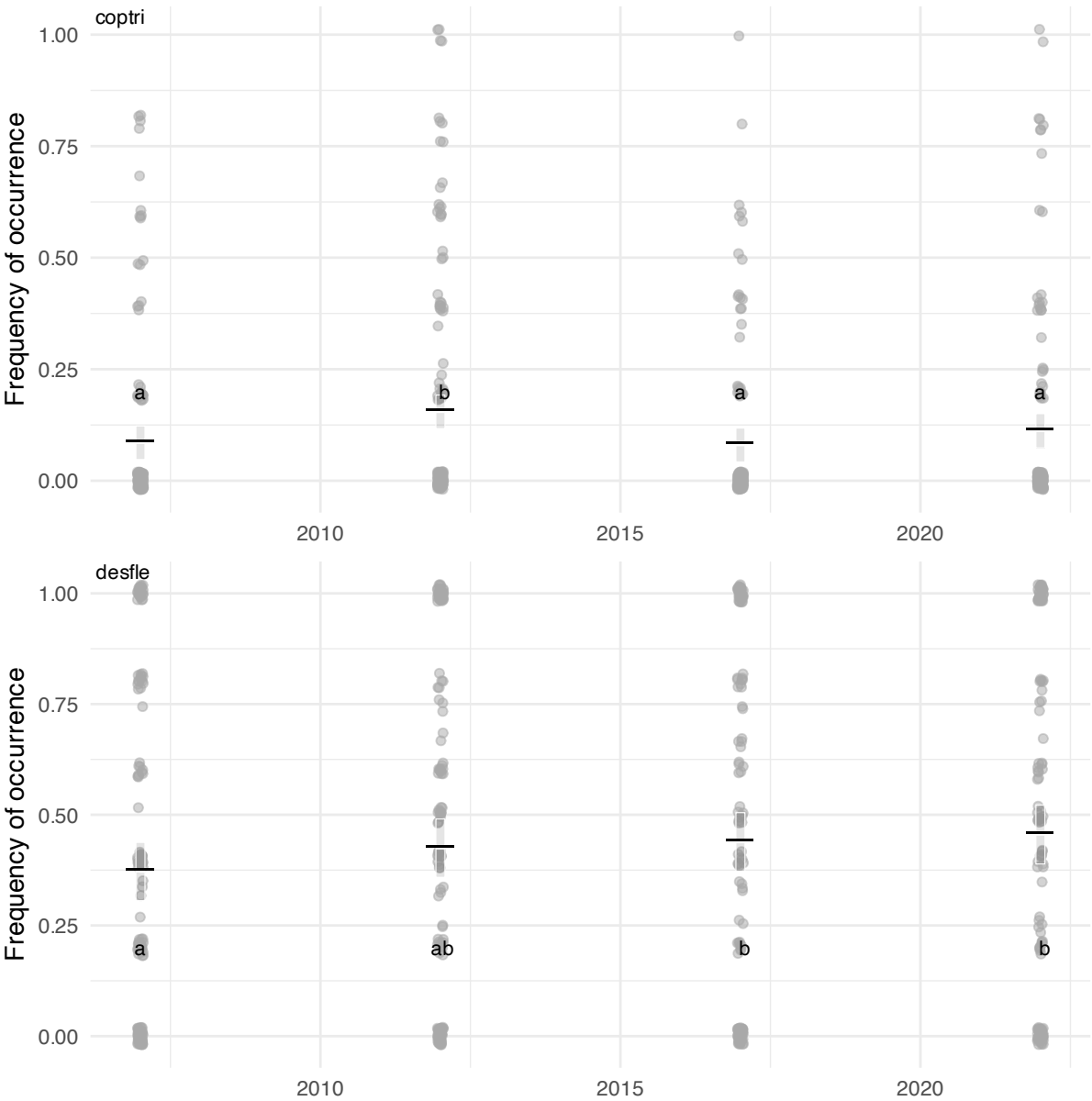
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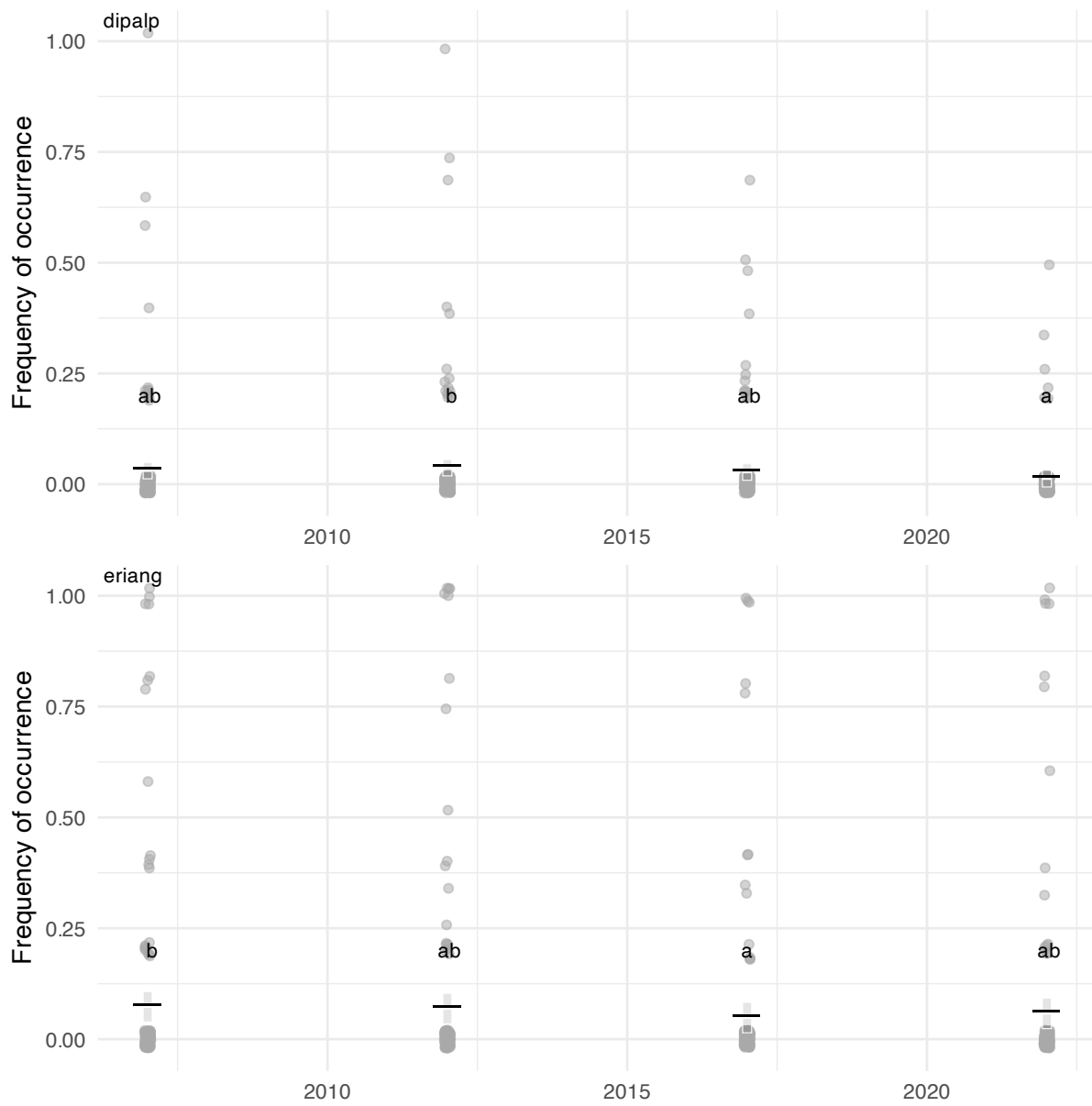
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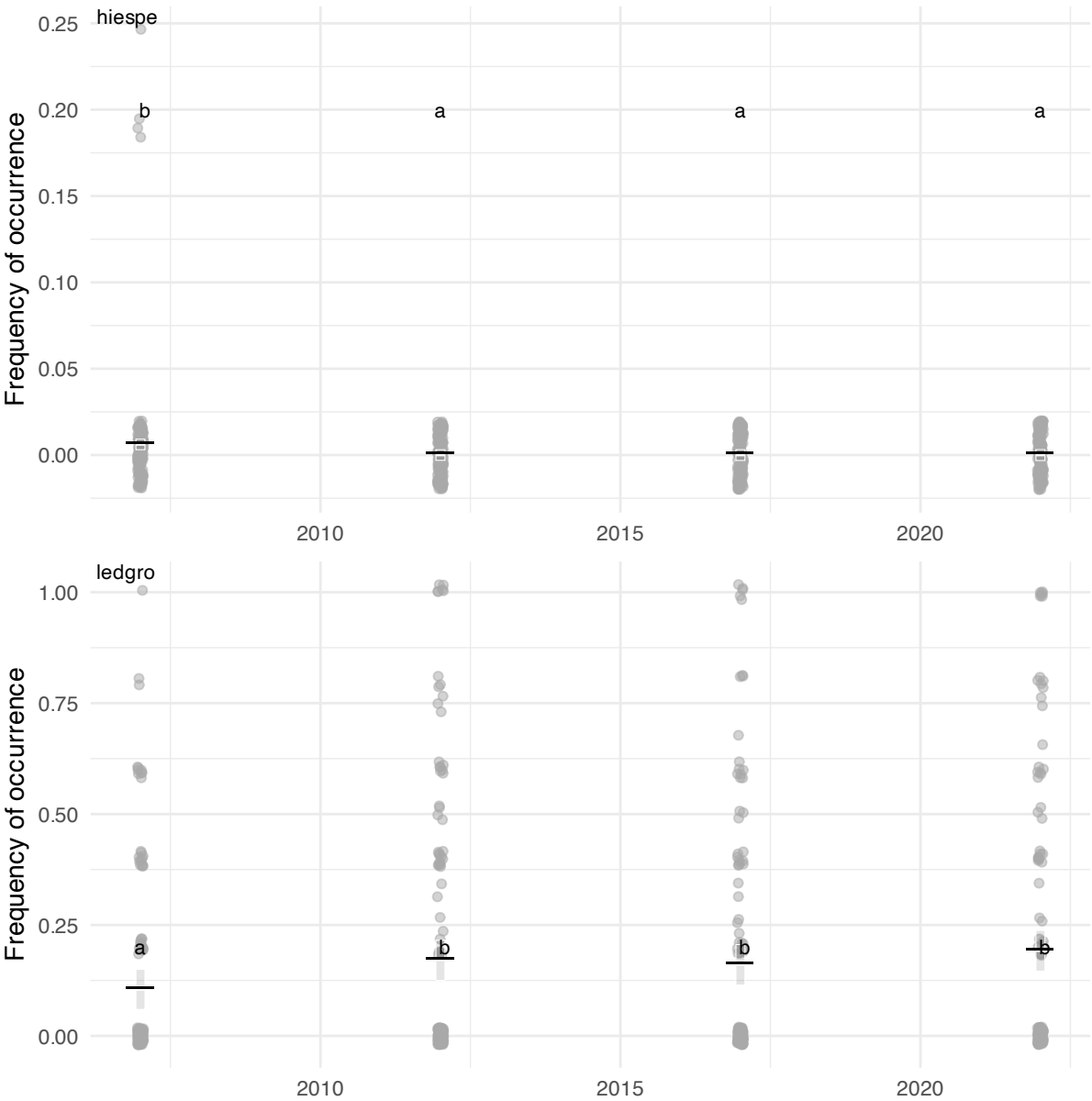
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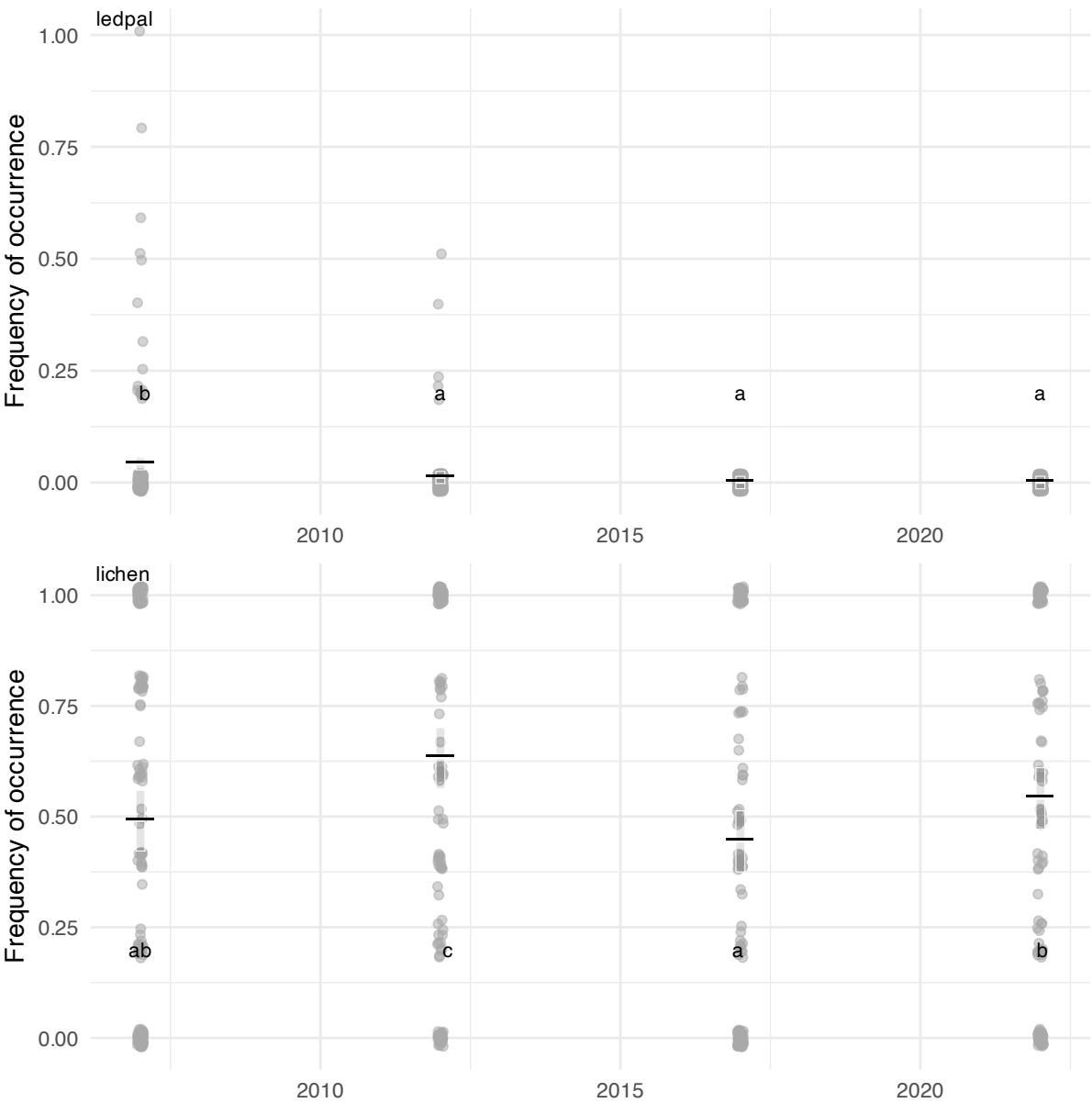


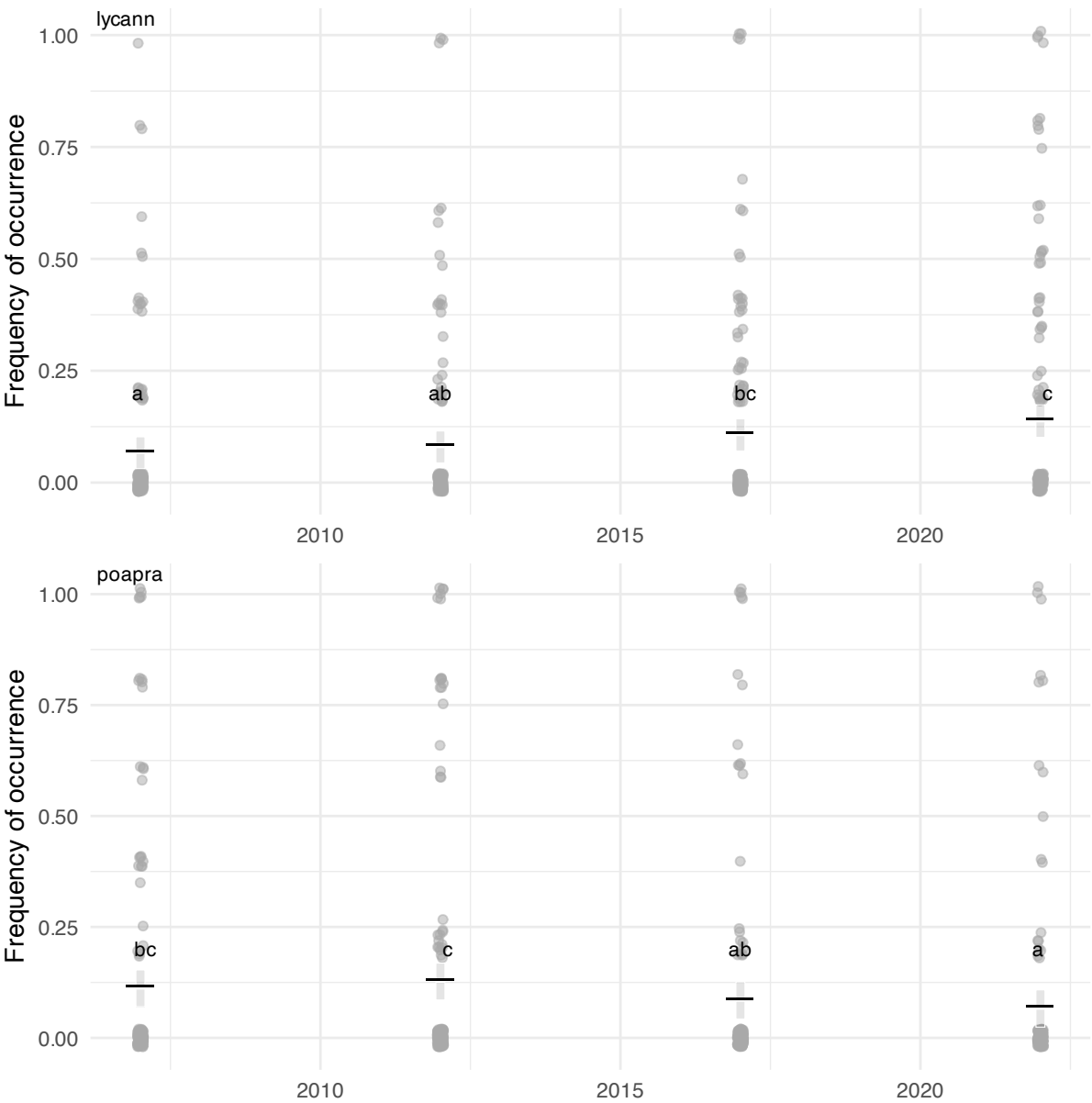


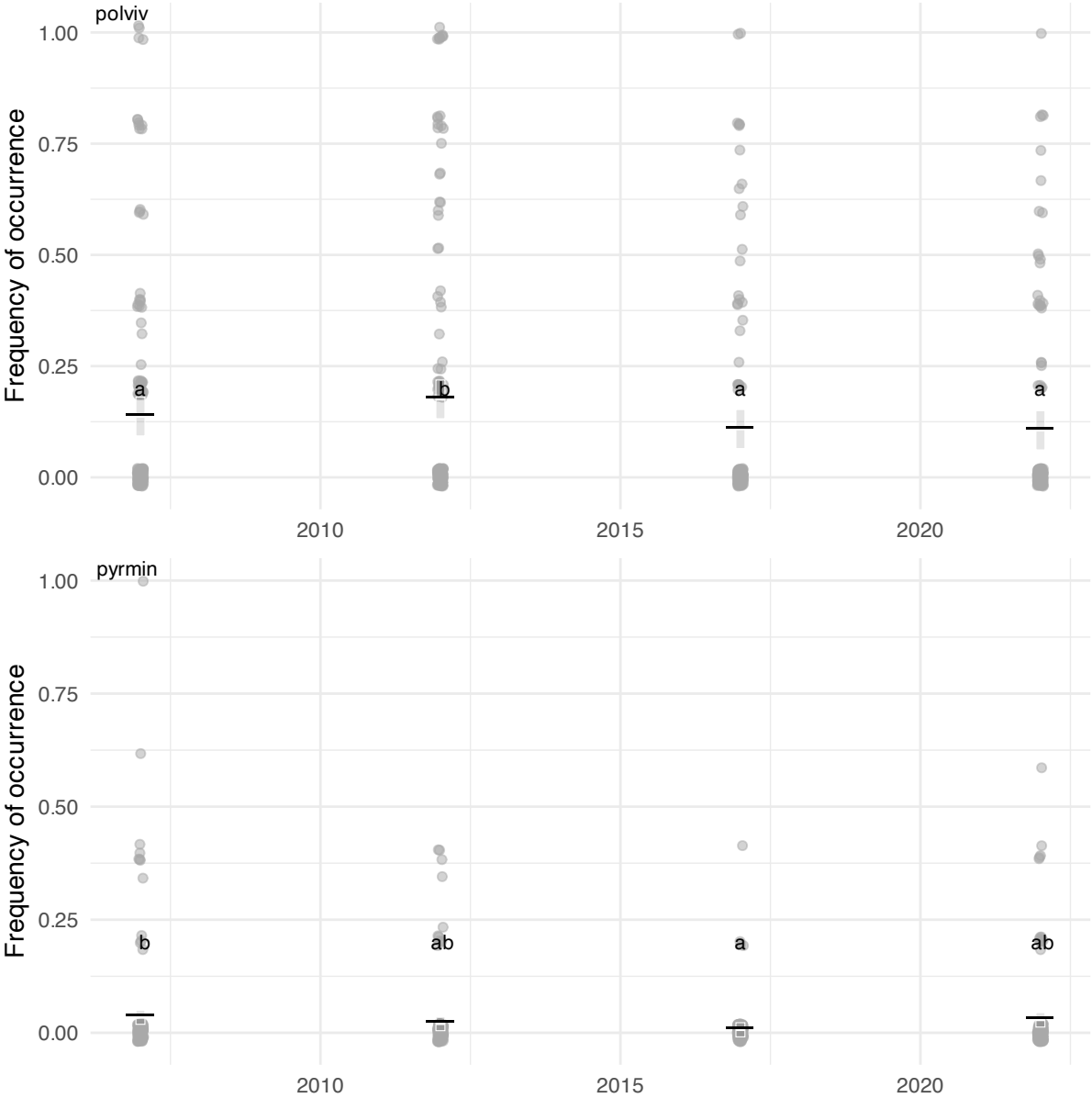


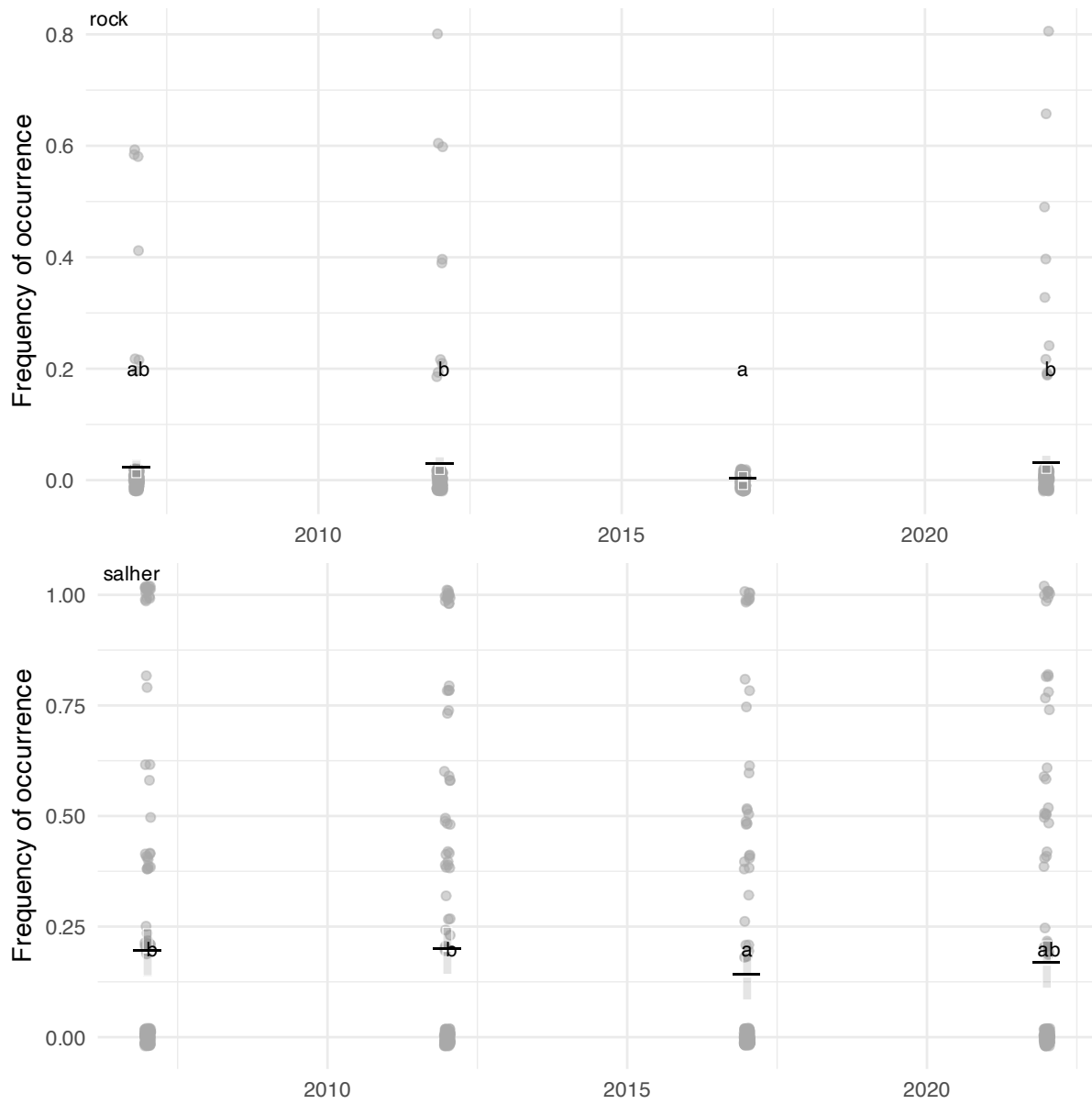


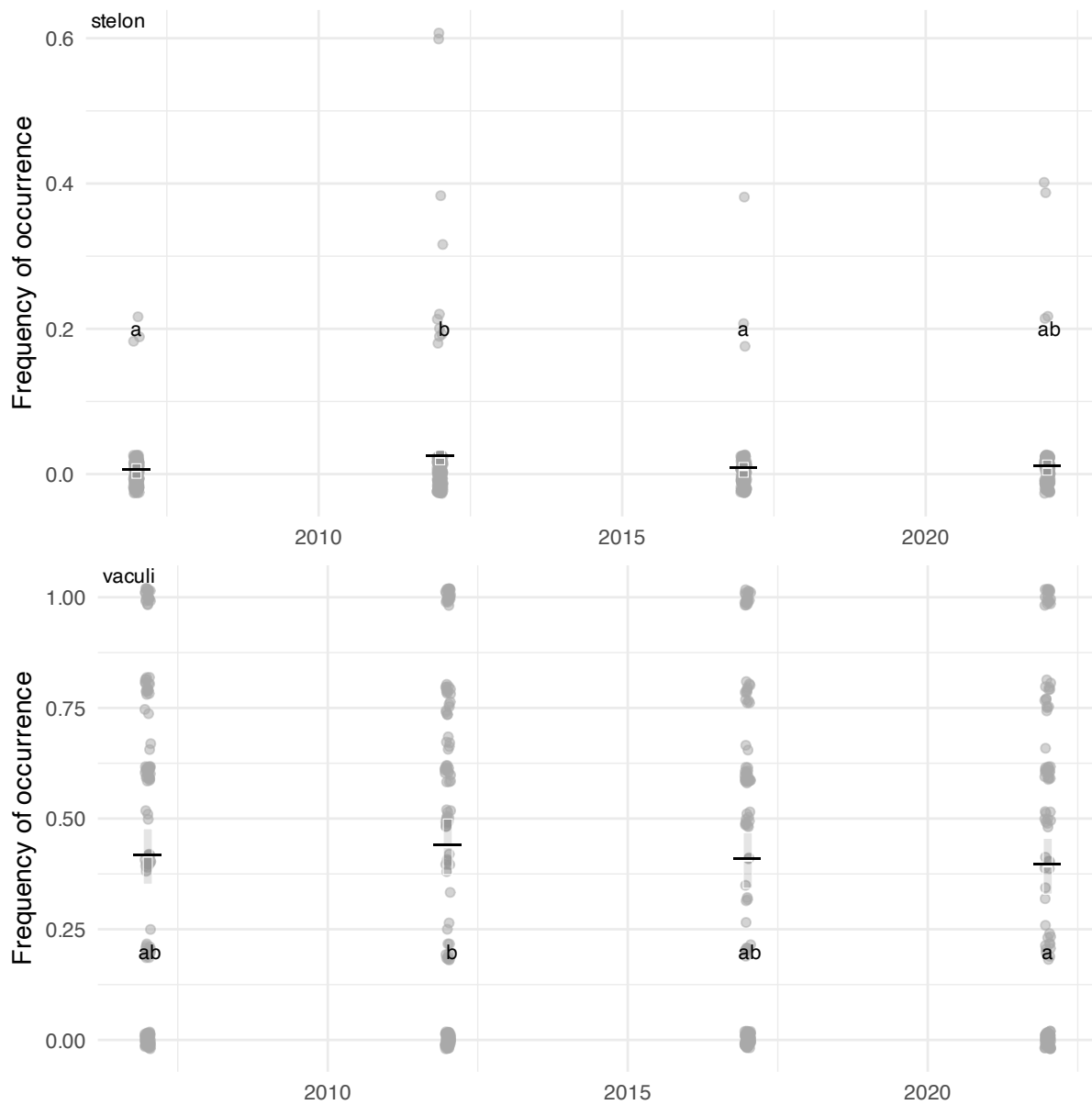












4 Phenology (discussion)

Data available from <https://data.g-e-m.dk> (GEM 2026b, 2026c).

4-1 NDVI

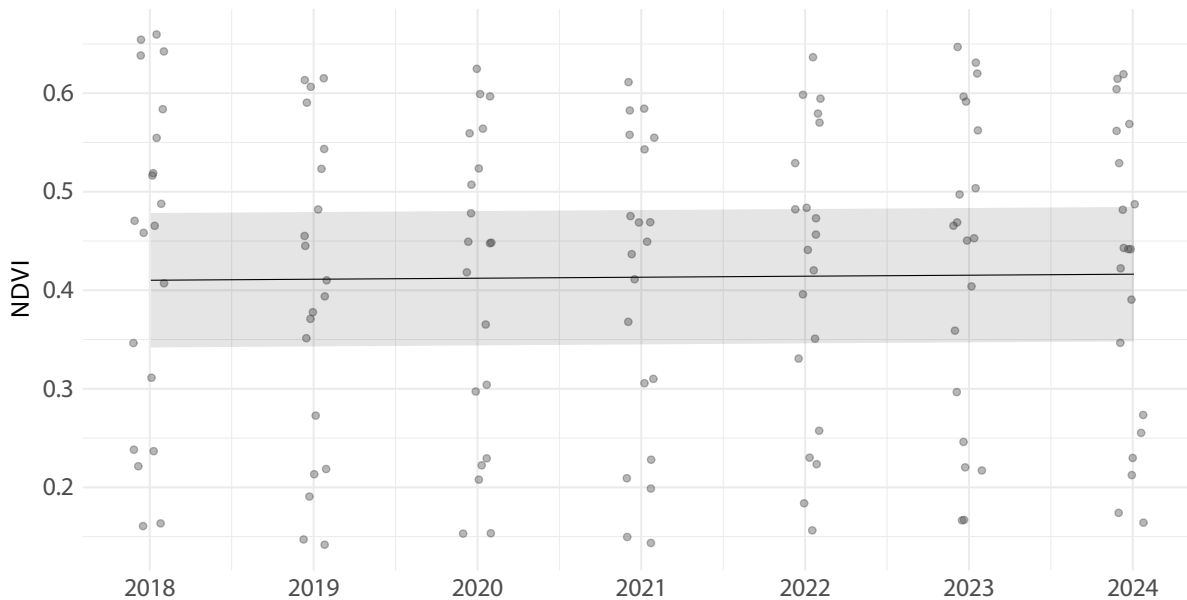


Figure 17: NDVI change over time. Mean NDVI per plot per year, based on in situ measurements from 2018 to 2024. The trend line shows a significant increase of 0.001 NDVI units per year (95% CI: $3\text{e-}04$ –0.0018), representing a cumulative increase of 0.0061 NDVI units over the study period.

5 References

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